CM 515: Data Visualization Module I

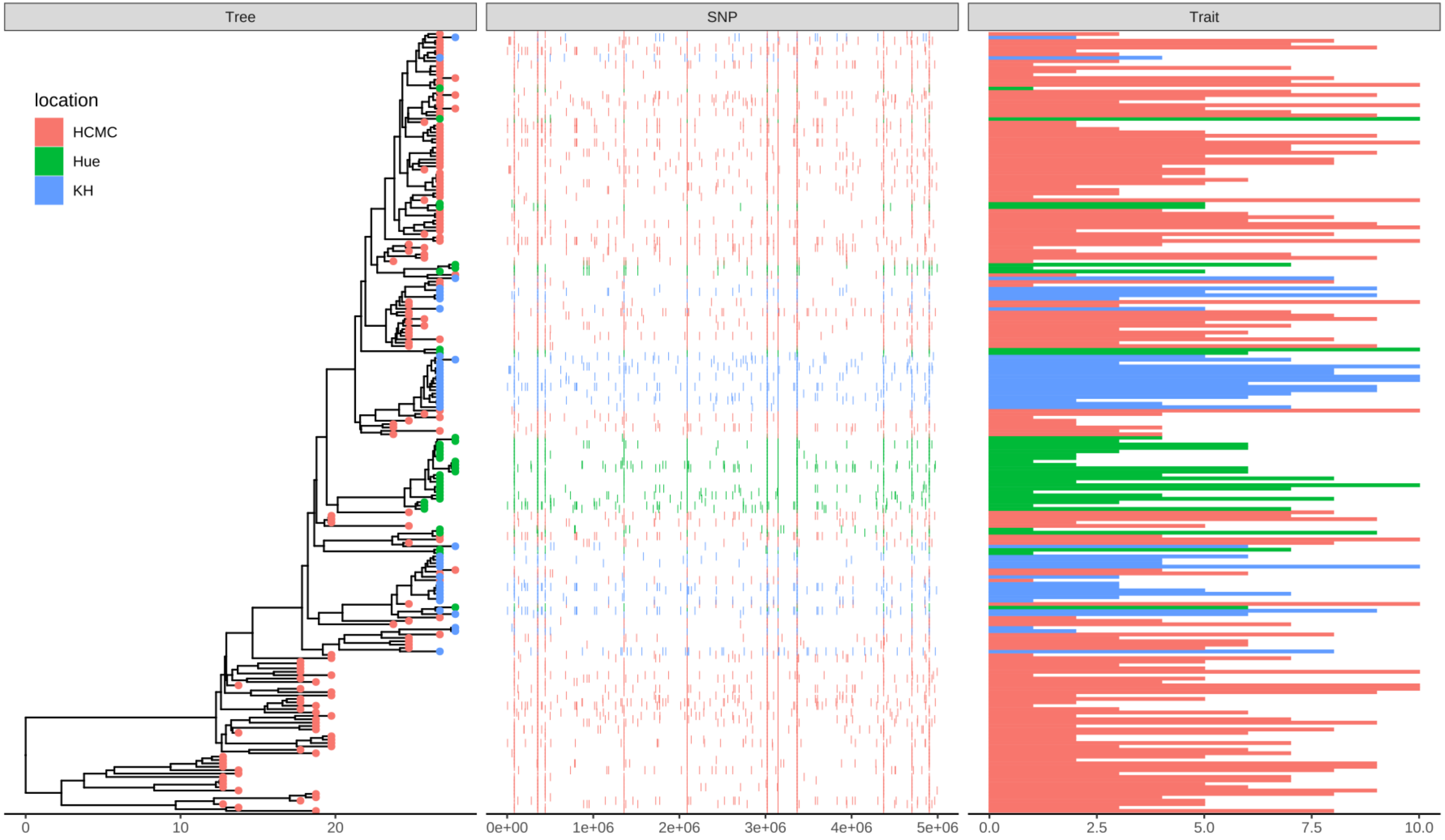

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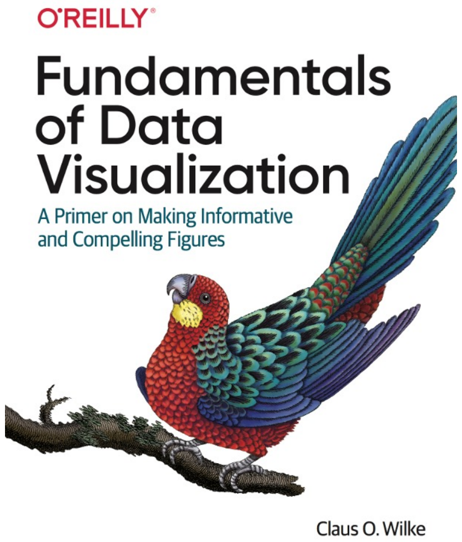
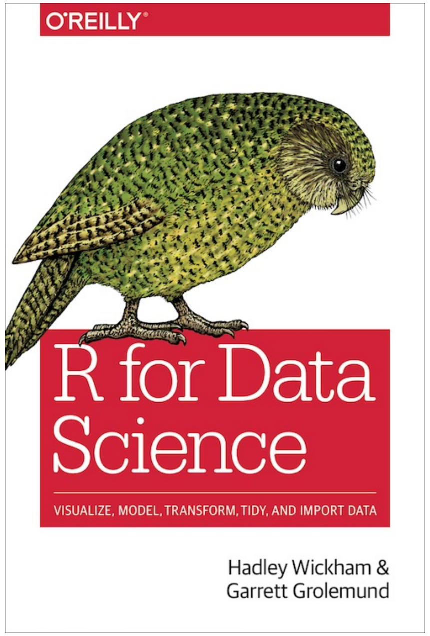
Image Source: Guangchuang Yu https://yulab-smu.top/treedata-book/chapter7.html

Data Visualization Resources

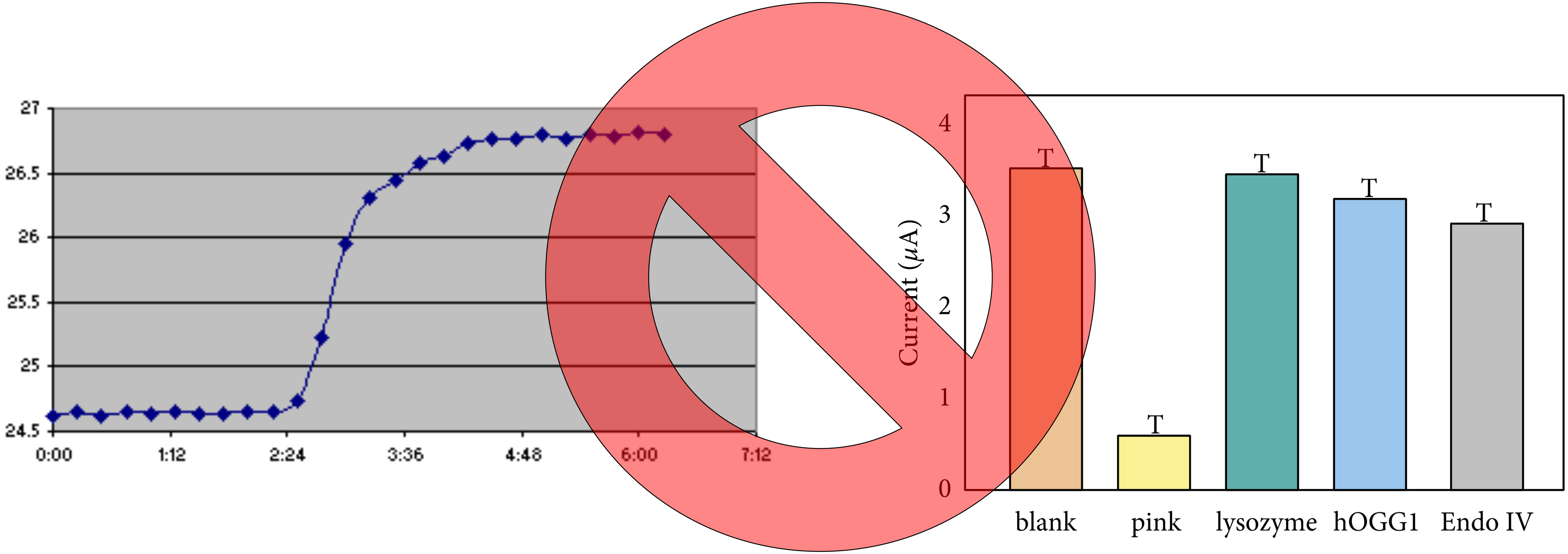
[Claus Wilke Data Visualization in R Course (U Texas)](#)

[R for Data Science](#)





Scientific Communication and Professionalism

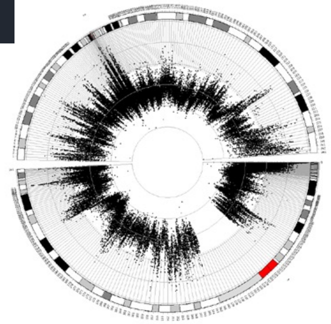


- Clear, accurate, and complete representation of your data
- Efficient, reproducible, and automated methods
- Clean, professional, and aesthetically pleasing appearance

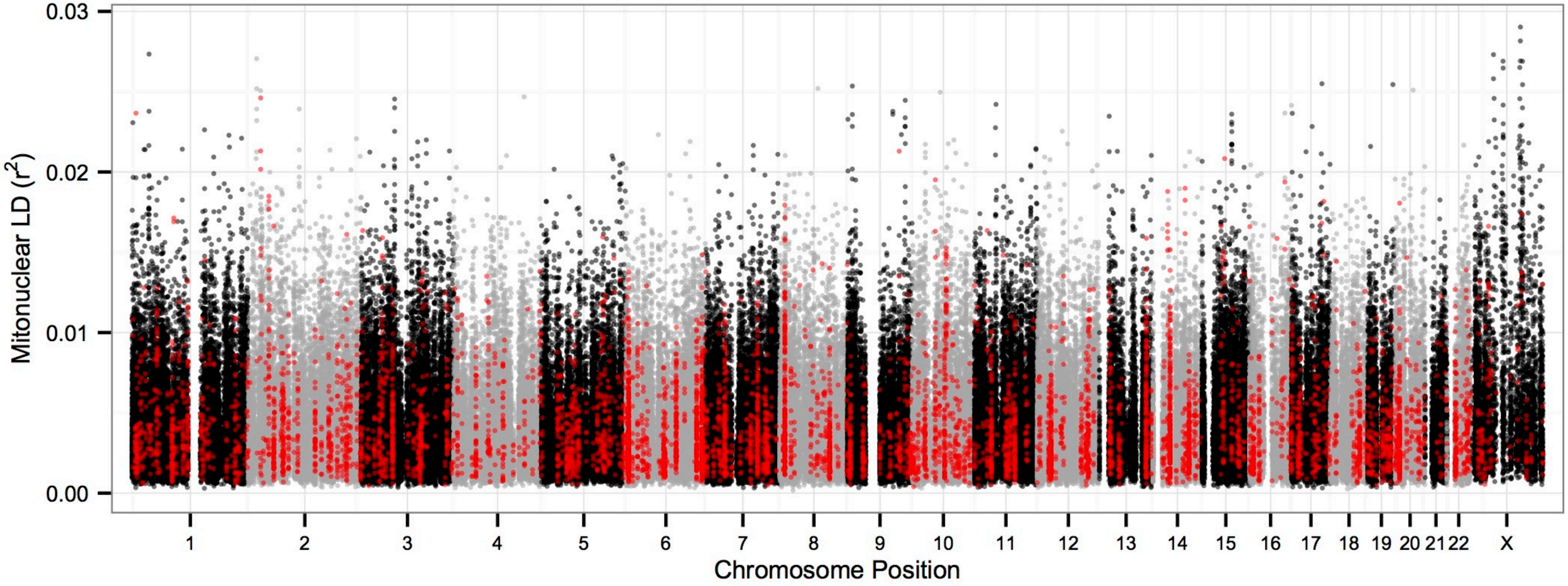
The Right Tools for the Job

A few examples

- R and ggplot
- Matplotlib
- Circos
- Processing
- Adobe Illustrator
- BioRender



Making Figures with Code




```
ggplot(cnld) +
        geom_point(aes(x=CumPos, y=r2, size=0.75, colour=as.factor(ChromPrint), alpha = 1/8)) +
        scale_size_identity() + theme_bw(base_size=15) +
        scale_color_manual(values=c(rep(c('black', 'dark gray'),11), 'black', 'red')) +
        scale_x_continuous(expand = c(0.015, 0.015),labels=c(as.character(1:chrNum), "X"), breaks=bpMidVec) +
        theme(plot.margin = unit( c(0.03,0.03,0.03,0.03) , "in" ), legend.position='none', axis.text.x =
                element_text(size=6), axis.text.y = element_text(size=7), axis.title.x =
                element_text(size=8), axis.title.y = element_text(size=8)) + xlab('Chromosome Position’) +
        ylab(expression(paste("Mitonuclear LD (",r^2, ")")))
```

The Grammar of Graphics

- **aes**: Aesthetic mapping of data to plot elements
  - position (X or Y coordinates), shapes, sizes, color, line weight/type, transparency, etc.
- **geoms**: Layers visually representing your mapped data
  - points, lines, bars, density curves, etc.
- **facets**: Dividing into subplots
  - partitions dataset based on one or more variables
- **themes**: Non-data plot elements
  - axis labels, grid lines, titles, etc.

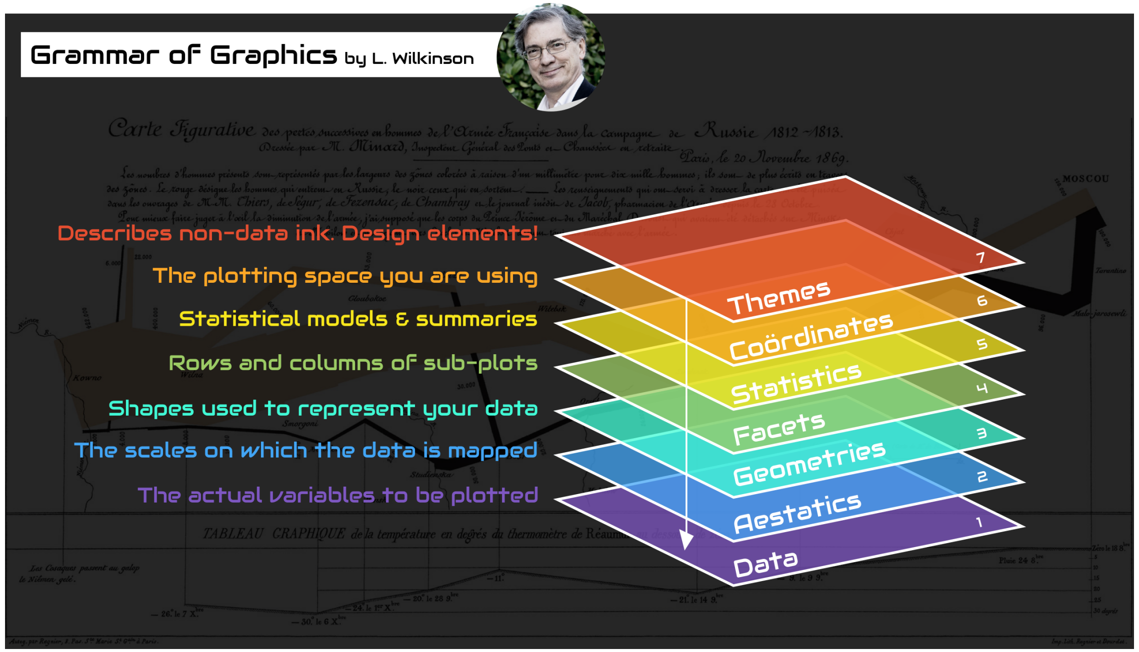


Image: Thomas de Beus

A Few Principles of Data Visualization

- What are you trying to communicate??
- The demise of the bar plot
- Choosing a scale: log vs. linear
- Use and choice of color palettes

Patterns

CellPress OPEN ACCESS

Perspective

Principles of Effective Data Visualization

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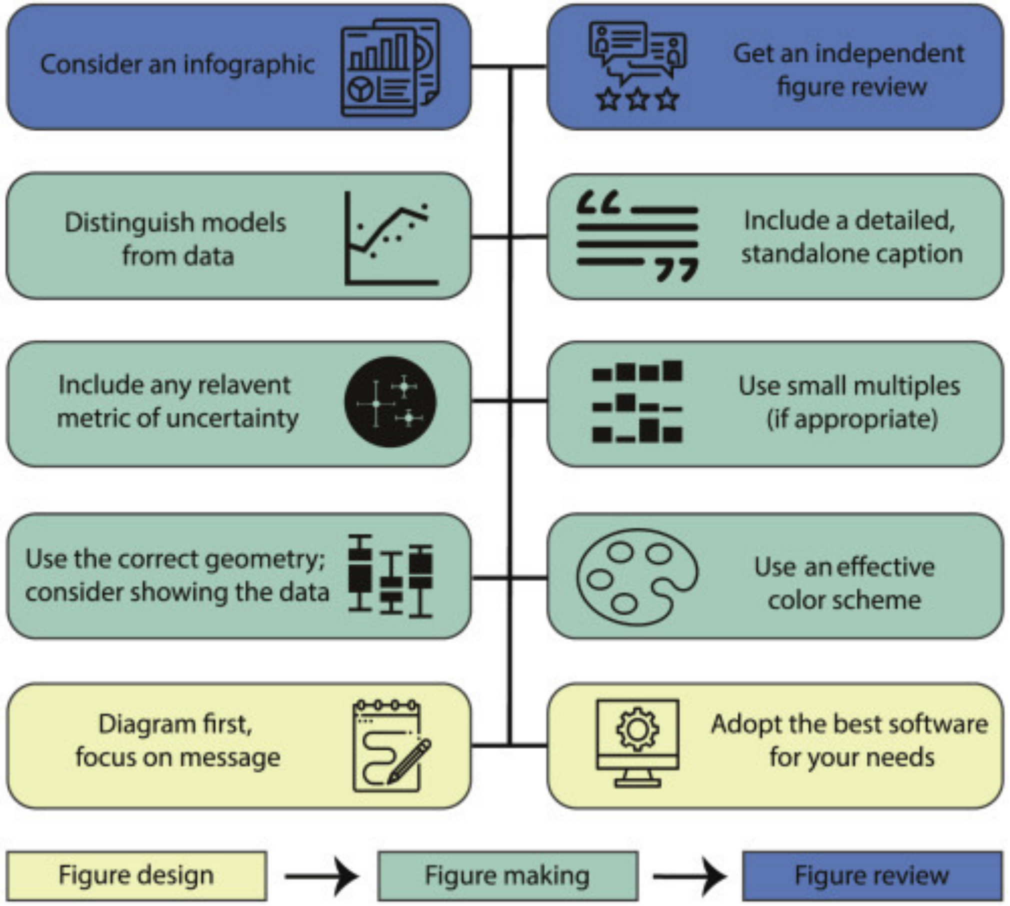
https://doi.org/10.1016/j.patter.2020.100141

https://www.sciencedirect.com/science/article/pii/S2666389920301896



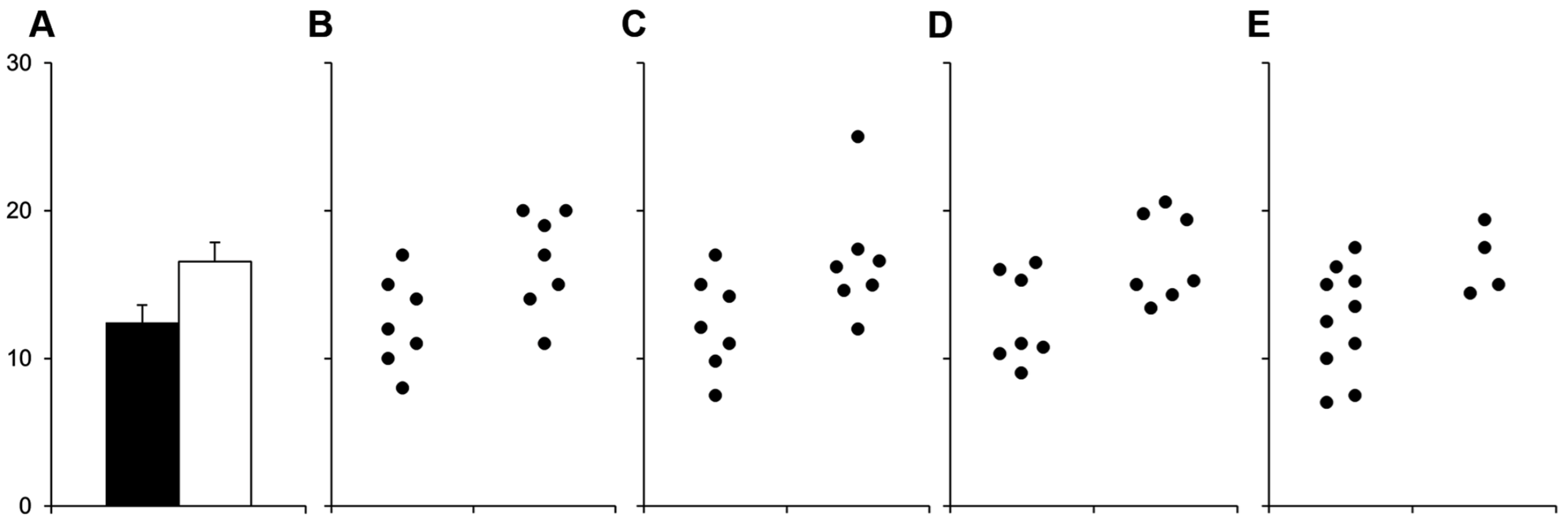
The Demise of the Bar Plot

PLOS BIOLOGY

Beyond Bar and Line Graphs: Time for a New Data Presentation Paradigm

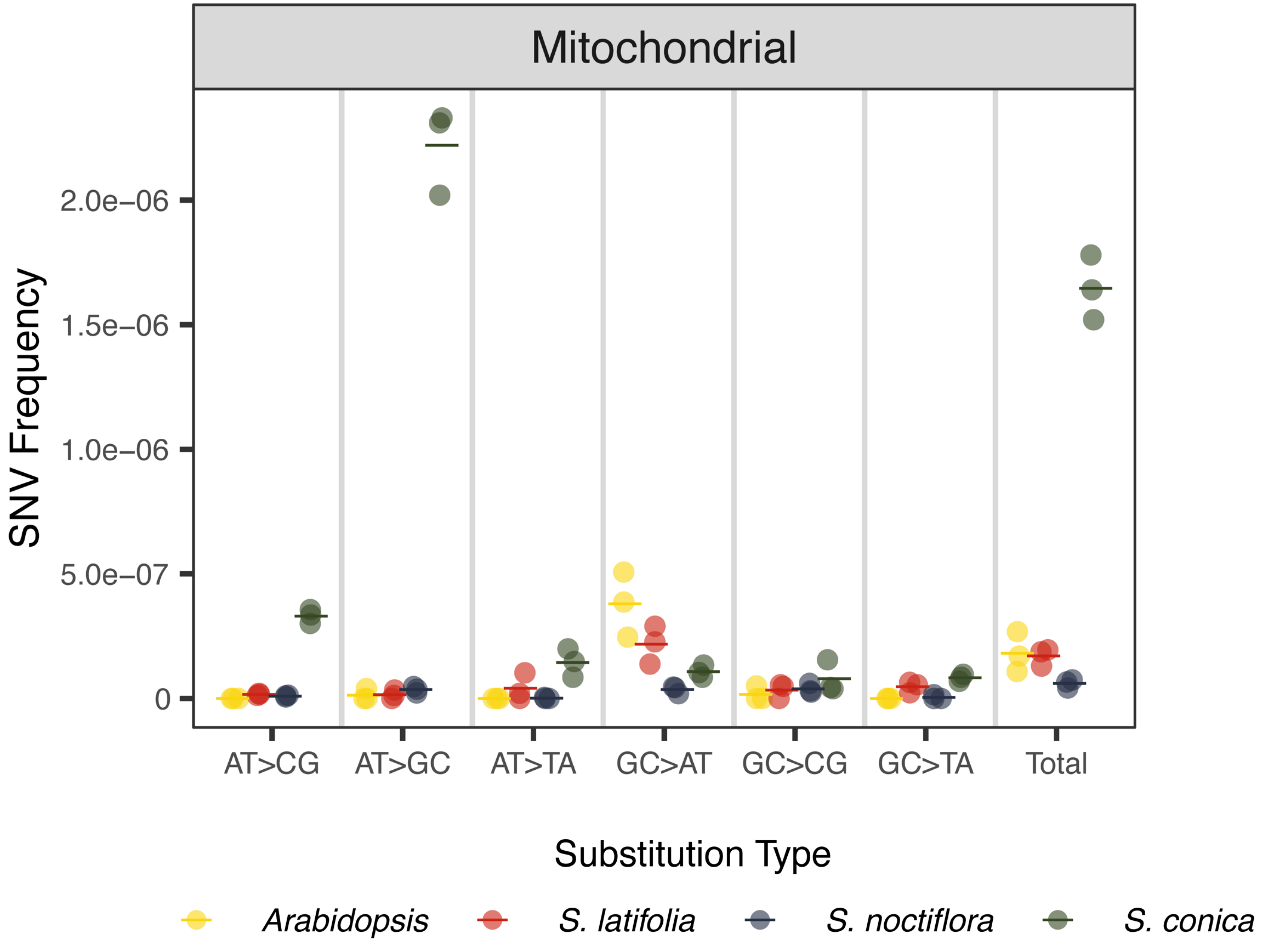
Tracey L. Weissgerber, Natasa M. Milic, Stacey J. Winham, Vesna D. Garovic

Published: April 22, 2015 • https://doi.org/10.1371/journal.pbio.1002128



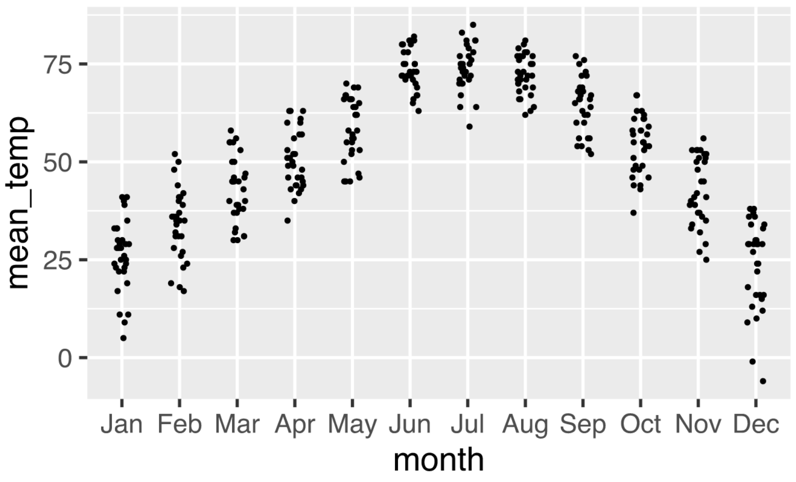
When Possible… Show All the Data!

Use point size, jitter, and/or transparency to mitigate the effects of overlapping points.

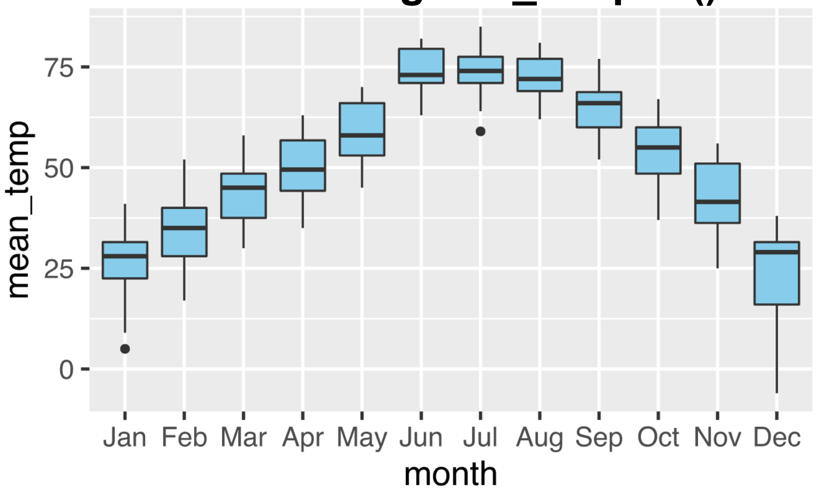


Better Ways of Comparing and Summarizing Distributions

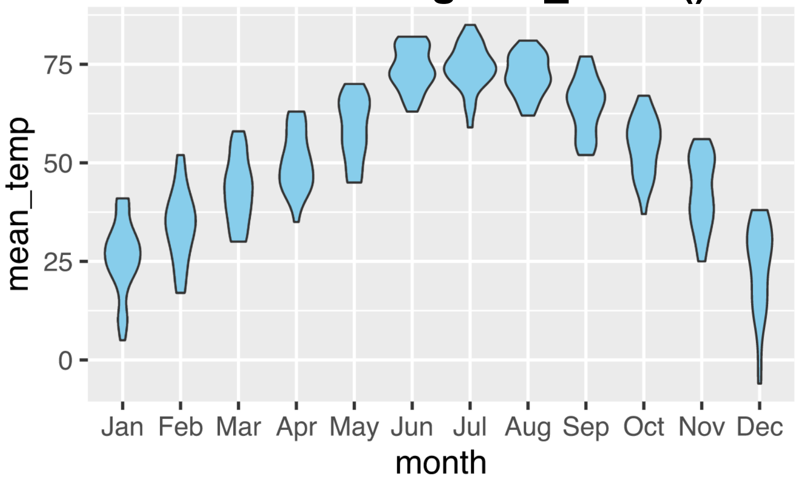
All Data Points: geom_point() [with jitter]



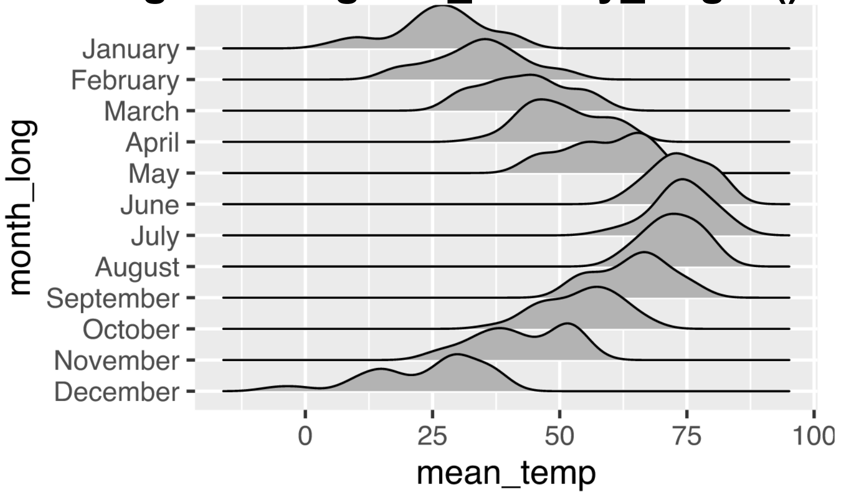
Box Plots: geom_boxplot()



Violin Plots: geom_violin()

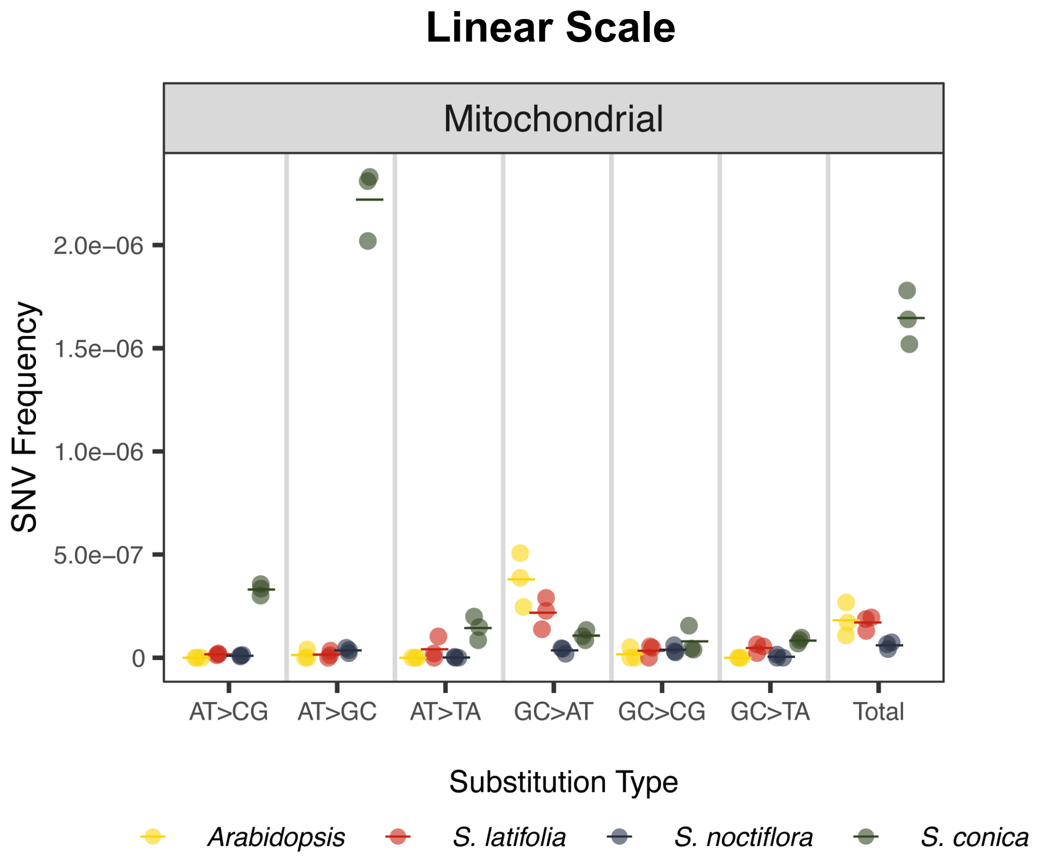


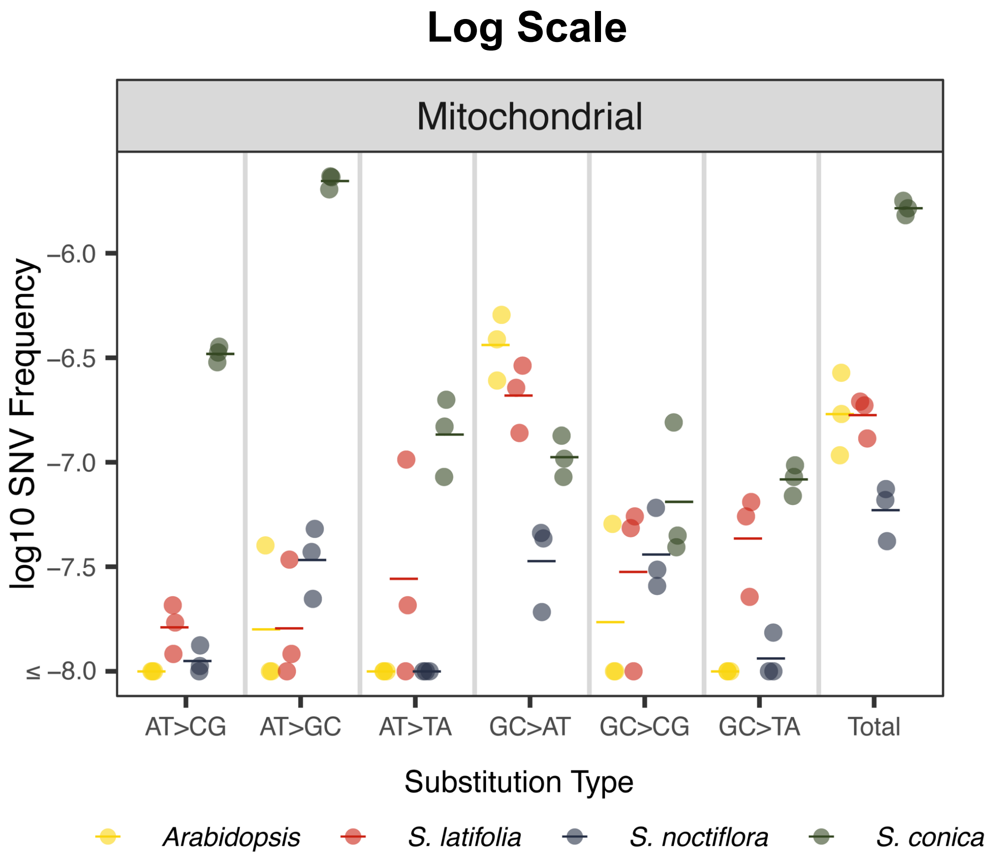
Ridge Plots: geom_density_ridges()



Examples from Claus Wilke: https://wilkelab.org/SDS375/slides/visualizing-distributions-2.html#1

Linear vs. Log Scales





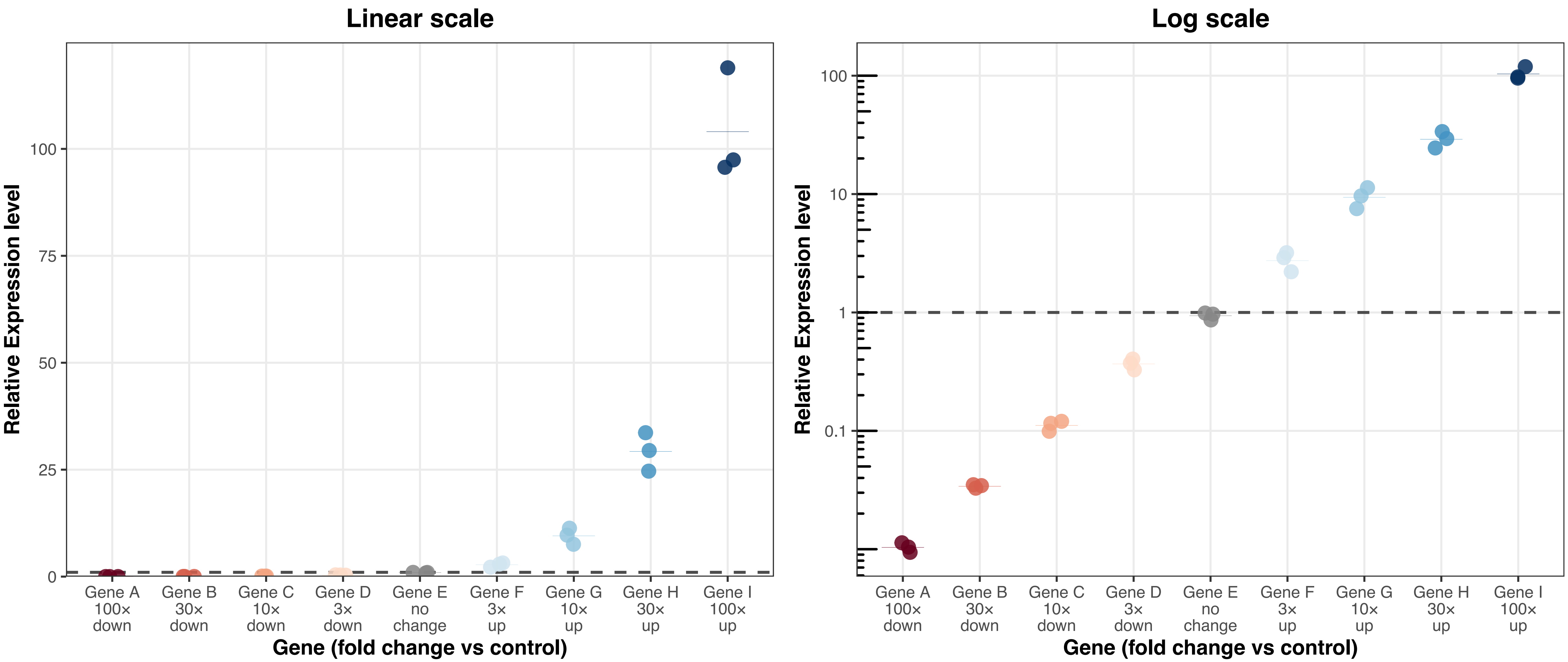
- Use **linear** scales to emphasize **absolute** differences.
- Use **log** scales to emphasize **proportional** differences.

Linear vs. Log Scales

RFK Jr: "A Democratic senator claimed it's mathematically impossible to have a drug drop by 600%. I said, 'Well, if the drug was $100 and it raises to $600, that would be a 600% rise. If it drops from $600 to $100, that's a 600% savings.'"

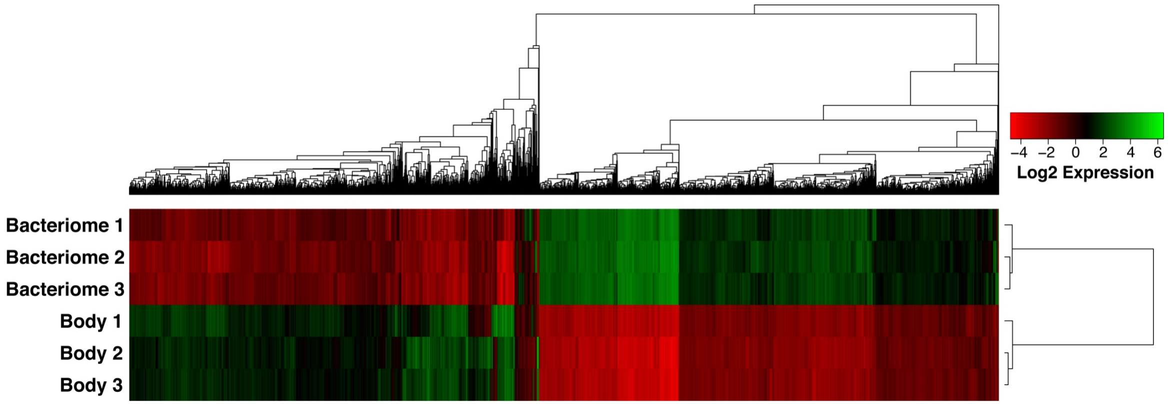
Trump: "Right"

Linear vs. Log Scales



Accessibility

Color is a powerful tool for visualizations, but it will not be perceived in the same way by everyone in your audience. Tips for making your visualizations accessible to color bind individuals….



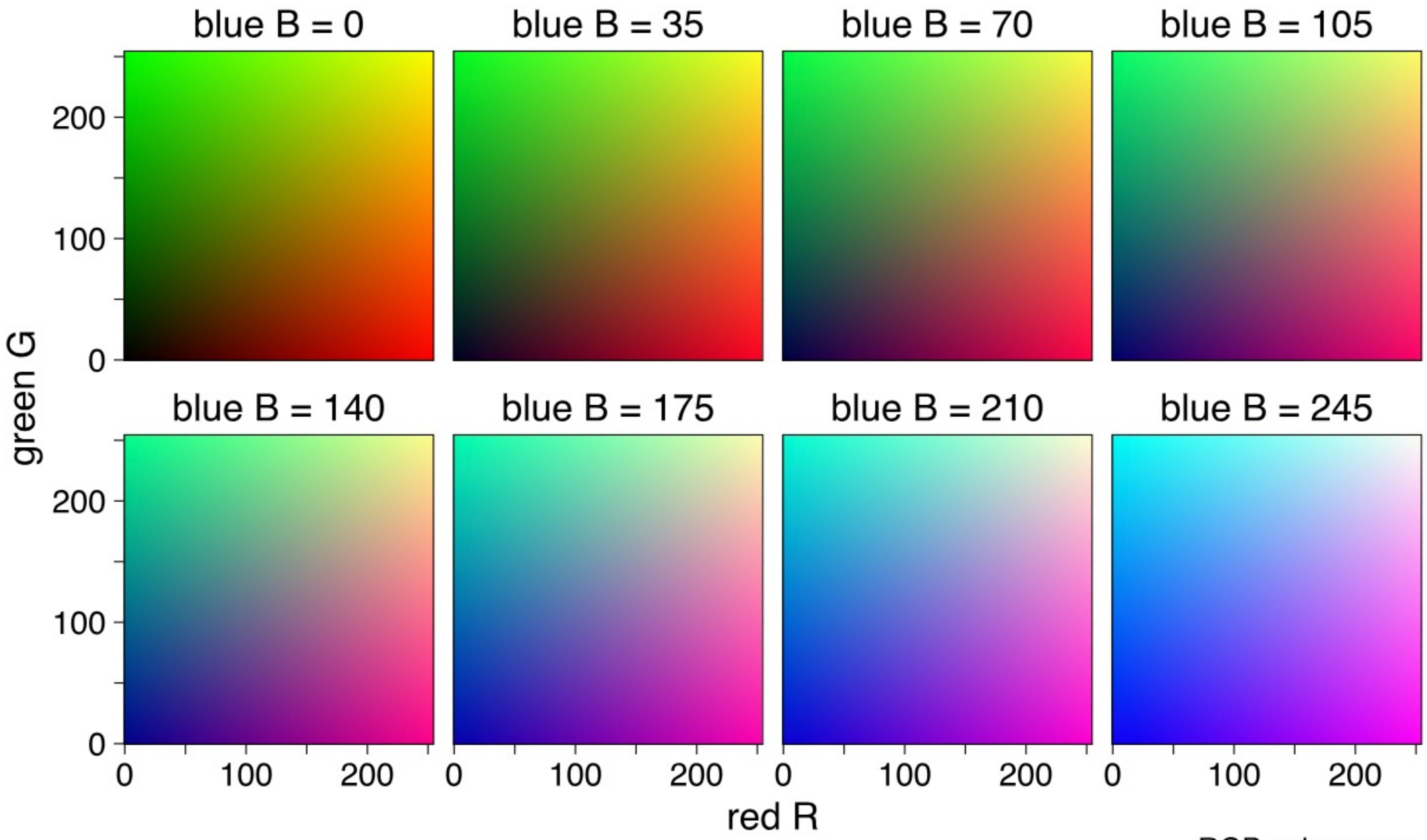
- Use [palettes consisting of colors that are more distinguishable]for individuals with common forms of color blindness.

- Use color and shape of points redundantly to distinguish among groups in plot.

```
> ggplot(MouseData, aes(x=age, y=weight, color=genotype, shape=genotype))
        + geom_point()
```

- Make use of figure labeling and legend descriptions to make the plot accessible even if colors are difficult to distinguish.

Choosing Colors

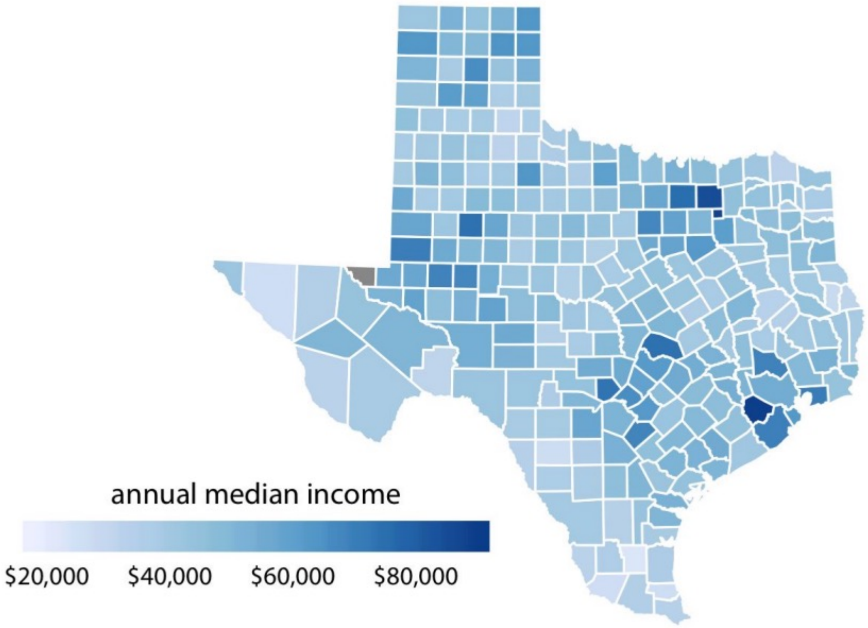


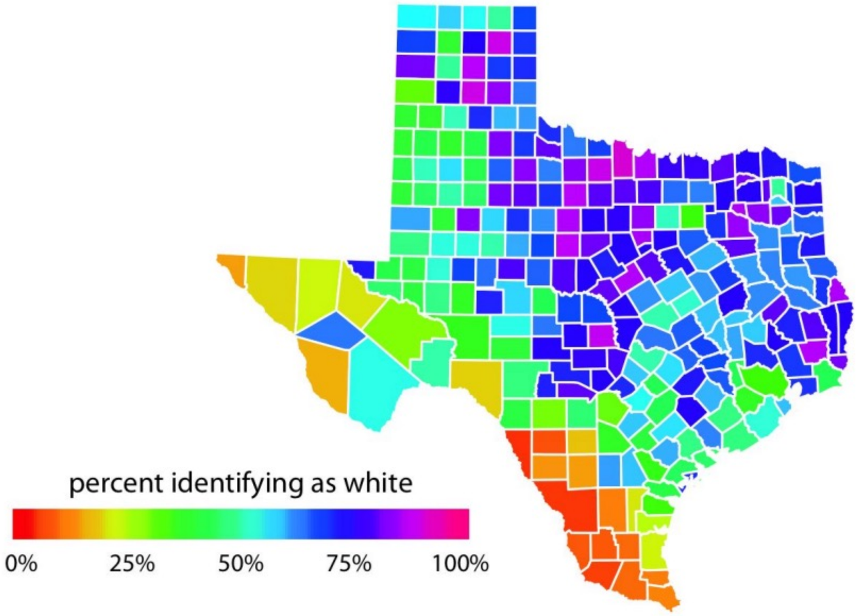
RGB color space

https://clauswilke.com/art/post/colors-color-spaces

Go Easy on Our Eyes

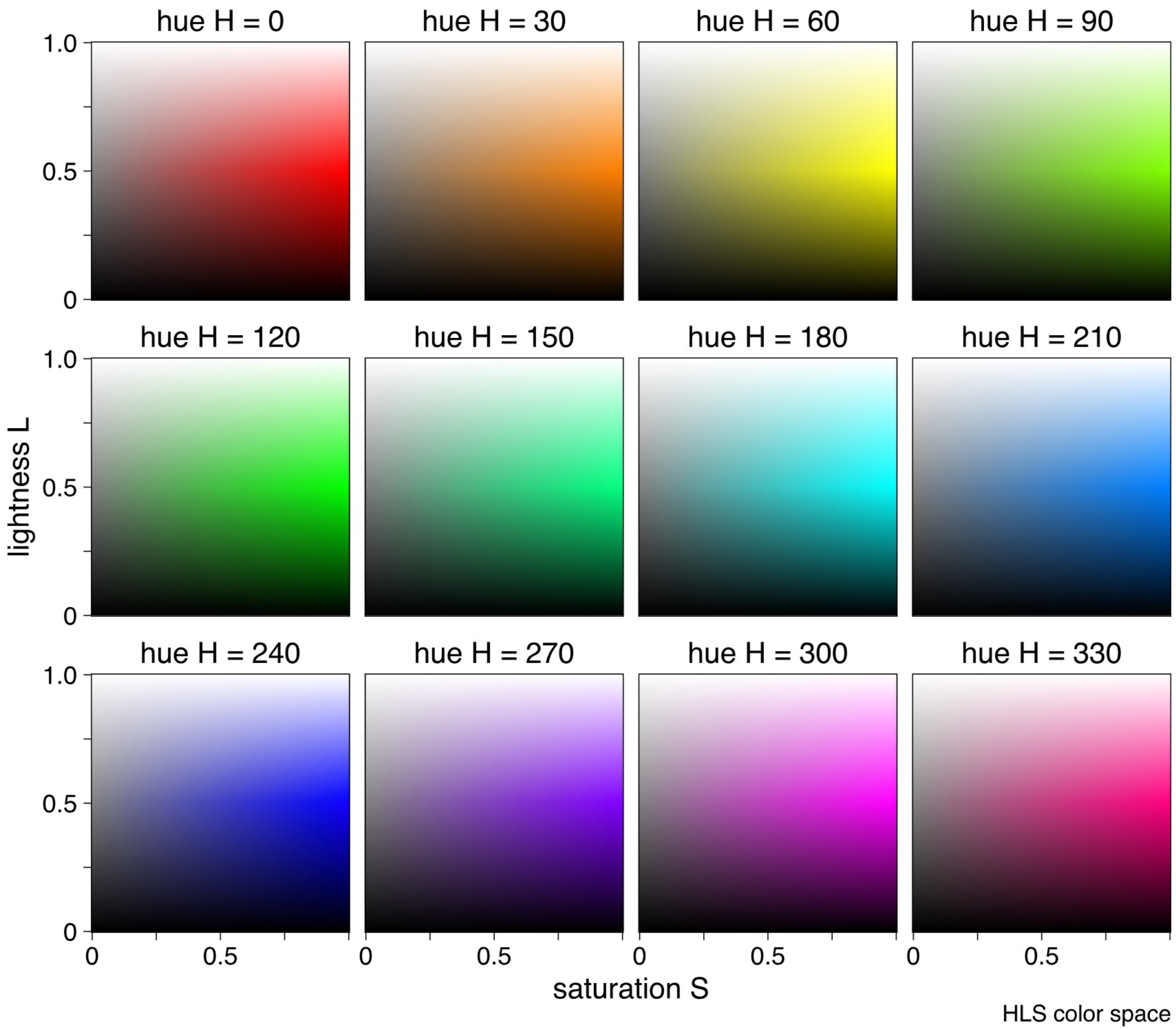
Generic named colors in R tend to have extremely high saturation.





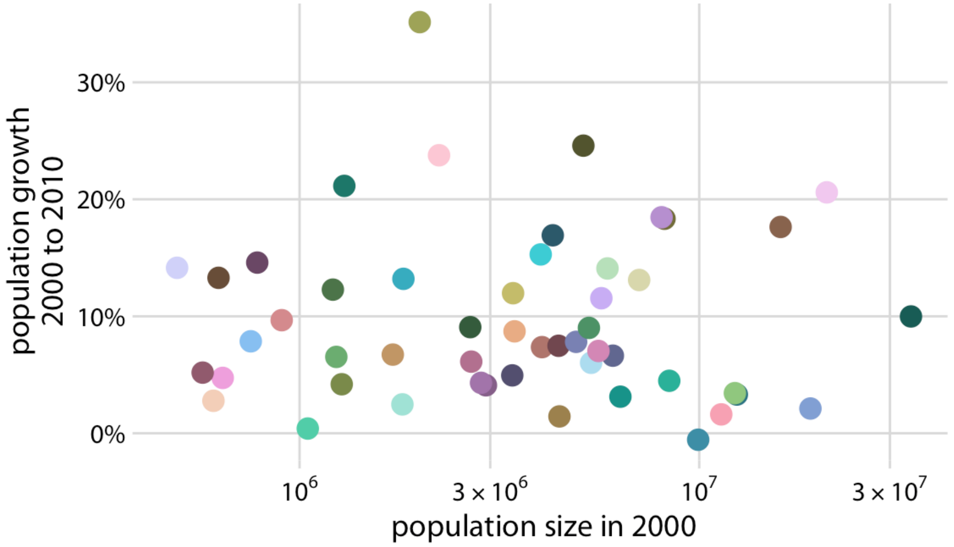
[Jess Cohen-Tanugi]()

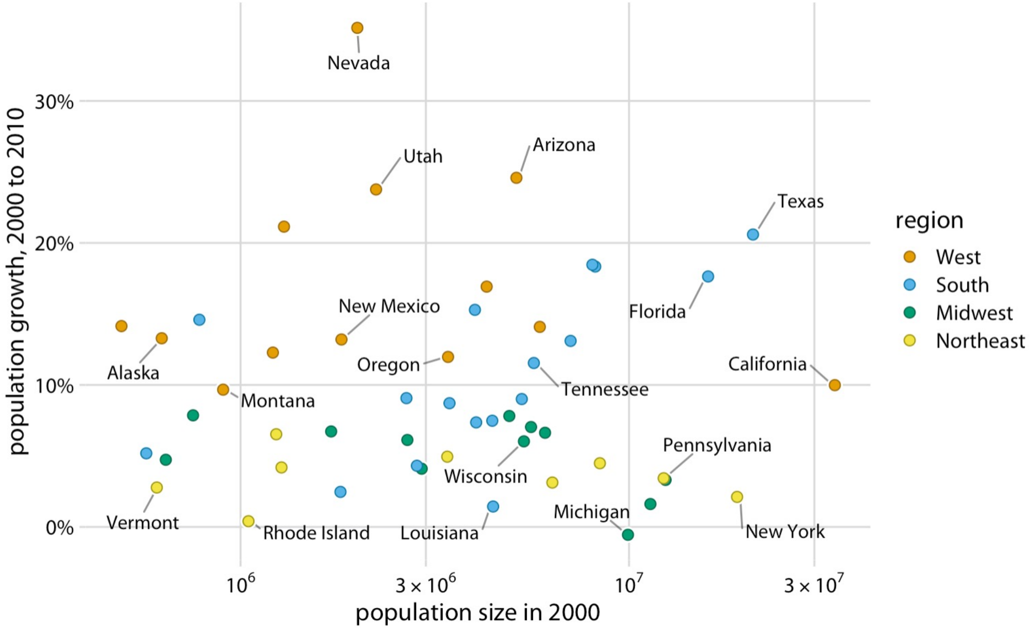
Thinking of Colors in Terms of Hue, Lightness and Saturation

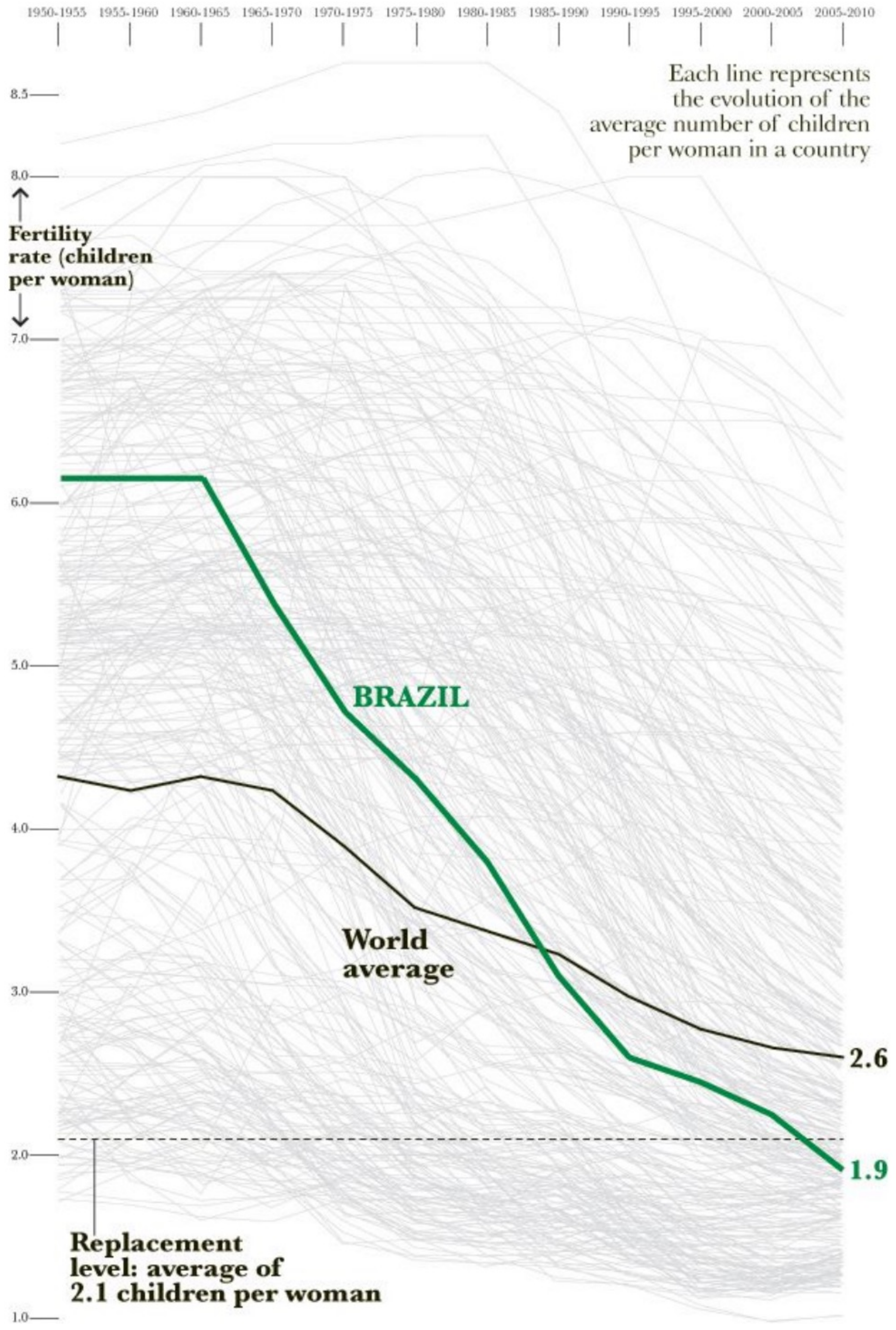


https://clauswilke.com/art/post/colors-color-spaces

Color to Emphasize, Not to Overwhelm







Jess Cohen-Tanugi

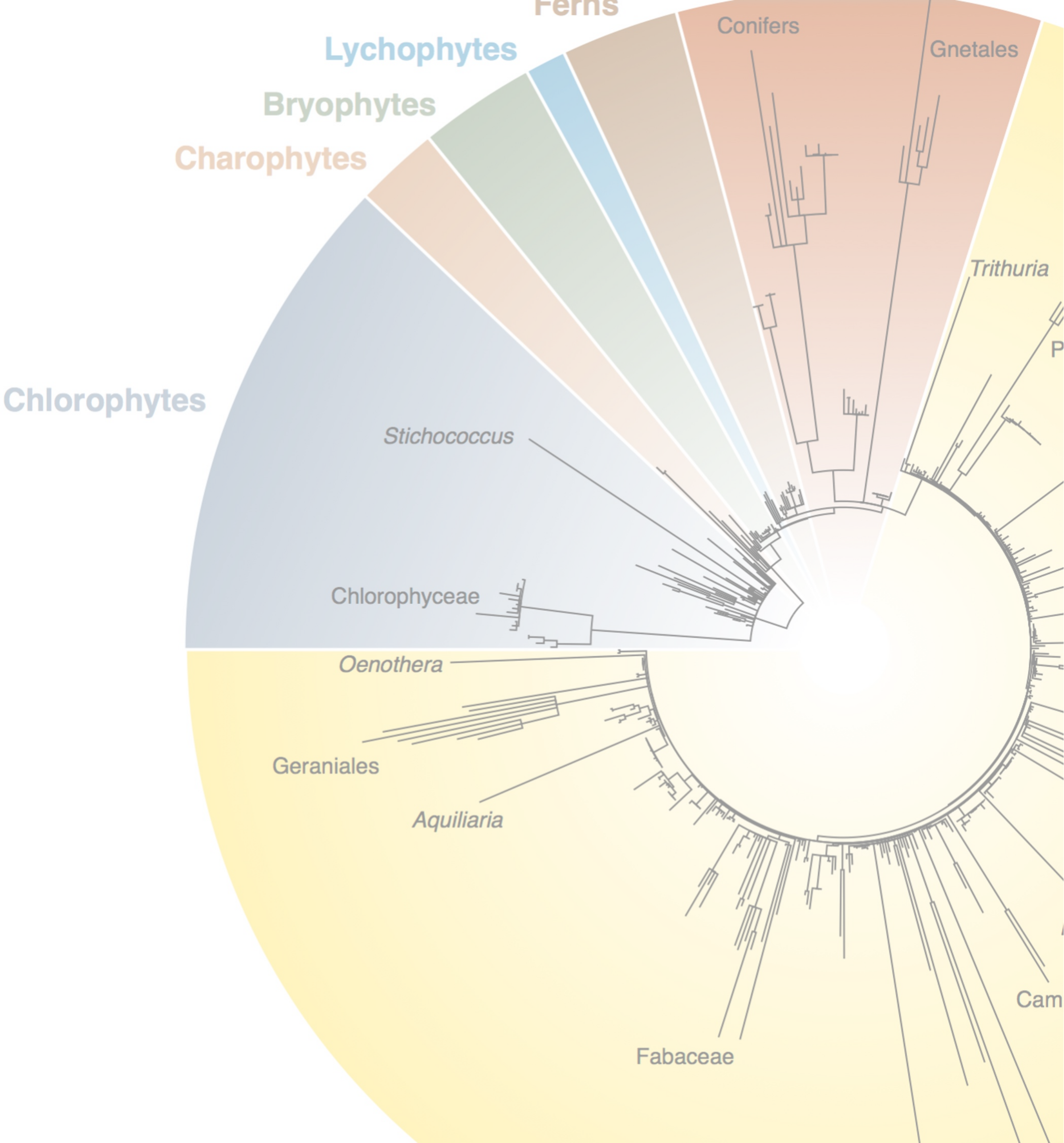
https://clauswilke.com/dataviz/color-pitfalls.html

Exercise and Assignment

https://dbsloan.github.io/CM515/SP26/ggplot/

Tree-Like Data Structures in Biology

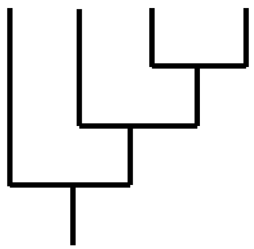
- The tree of life and species relationships
- Gene family evolution
- Hierarchical clustering of gene expression patterns, ecological/microbiome communities, etc.



Newick Tree Format

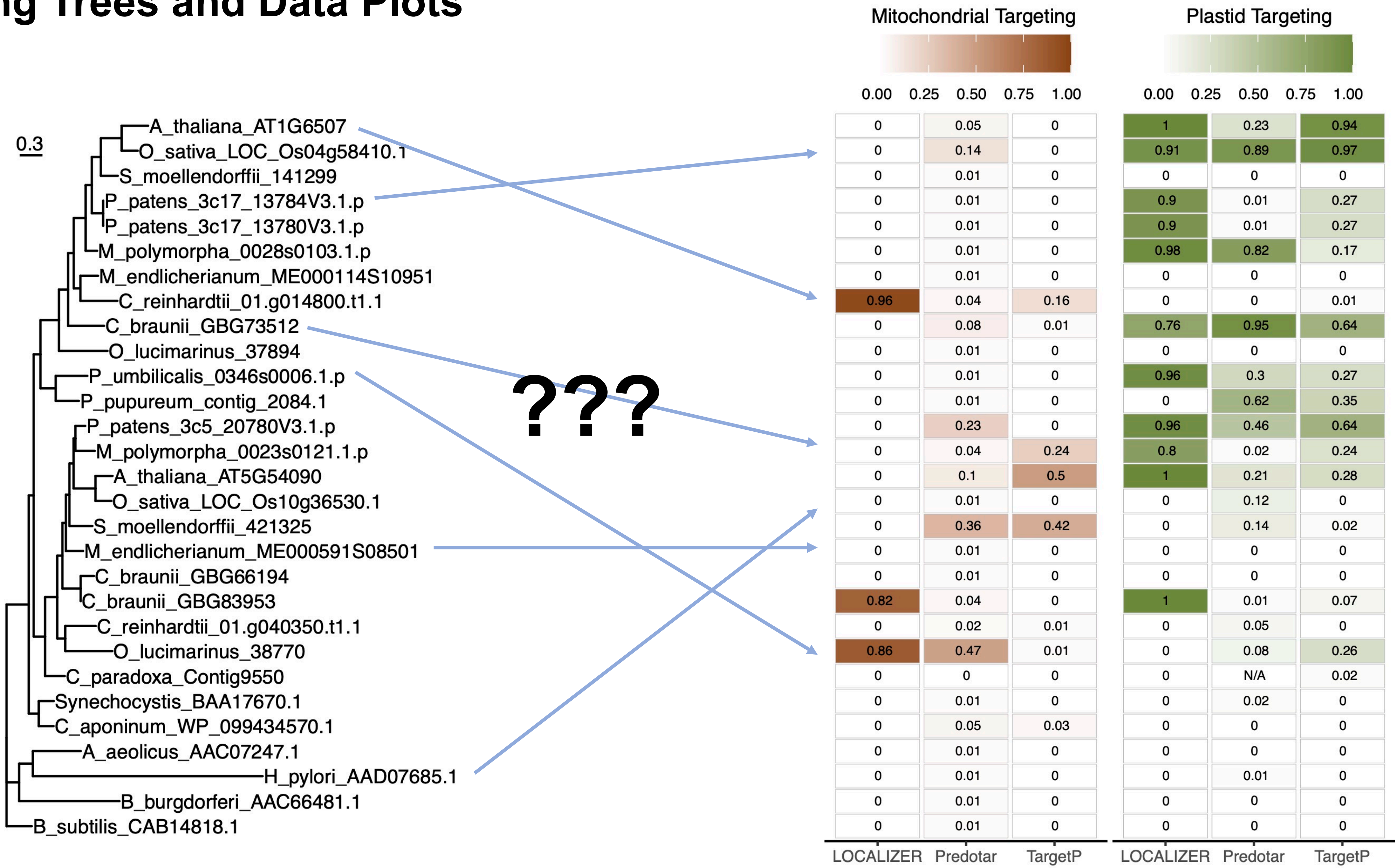
- Standard parentheses/comma-based format for summarizing tree branching patterns

(A, (B, (C, D)));



- Numerical values and annotations can be added to record features such as branch lengths, statistical (e.g., bootstrap) support, node/branch labels, and other features.

Linking Trees and Data Plots



Exercise and Assignment

https://dbsloan.github.io/CM515/SP26/ggtree/